



Day 7 — Cloud Storage Solutions

ITNW 1009 · AWS Academy Labs

Today's session delivers two complete, hands-on laboratory guides covering Amazon Elastic Block Store (EBS) and Amazon S3 with Glacier Deep Archive lifecycle policies. You will build career-ready cloud storage skills in both a guided and unguided AWS Academy environment.

LAB PART A: GUIDED EBS LAB

LAB PART B: SANDBOX S3 & GLACIER LAB

Today's Learning Agenda



Lab Part A — Guided EBS Lab

Work within the AWS Academy pre-packaged lab to create, attach, format, and snapshot an EBS volume on a live EC2 instance.



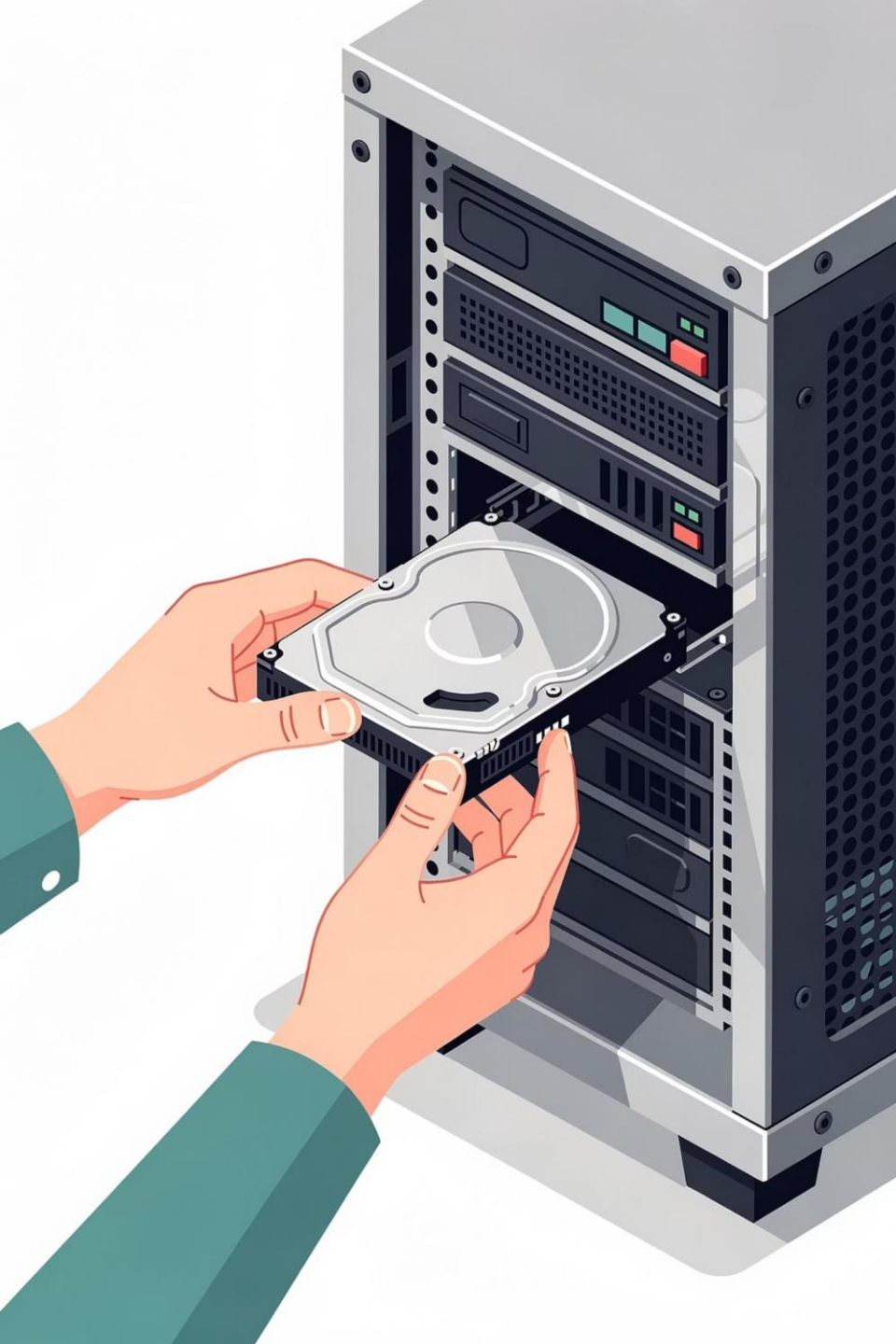
Lab Part B — Sandbox Scratch Build

Act as a Lead Cloud Engineer: build a VPC from scratch, deploy an EC2 host, create an S3 bucket, and configure an automated Glacier Deep Archive lifecycle policy.



Troubleshooting & Cleanup

Review common errors, remediation steps, and the mandatory environment teardown sequence to protect your \$100 sandbox budget.



LAB PART A

Working with Amazon Elastic Block Store (EBS)

Amazon EBS provides persistent, high-performance raw block storage that behaves exactly like a physical internal hard disk attached to your server. In this 75-minute guided lab, you will gain critical, career-ready skills in provisioning, formatting, mounting, and backing up EBS volumes on cloud compute instances.

Part A — Lab Objectives

By the end of this guided lab segment, you will be able to:

1

Create an EBS Volume

Provision a standalone 10 GiB General Purpose SSD (gp3) EBS volume from the AWS Management Console.

2

Attach to EC2

Attach the volume to a running EC2 instance within the same Availability Zone.

3

Format & Mount

Access the Linux command line to create an XFS file system, set up a mount point, and verify disk capacity.

4

Snapshot Backup

Execute a point-in-time EBS snapshot to ensure enterprise-grade data durability and complete lab grading checks.

Prerequisites & Critical Permission Rules

What You Need Before Starting

- **Active Environment:** Pre-packaged AWS Academy Guided Lab template (Lab 4: Working with EBS)
- **Linux CLI Familiarity:** Basic comfort with shell commands: `lsblk`, `mkfs`, `mount`, and `df`

Critical Permission Rules

AWS Academy environments are governed by **Service Control Policies (SCPs)**. Violating these triggers immediate "Access Denied" errors.

- Do **NOT** create custom IAM Users, IAM Groups, or root billing notifications
- Always use the pre-provisioned security profile named **LabRole** when authorizing instances or writing policies

Task 1 — Initialize the AWS Pre-Packaged Environment

1 Open the Lab Portal

Log into your **AWS Academy Learning Portal** (Canvas). Locate **Lab 4: Working with EBS** in the navigation links and click it.

2 Start the Lab

Click the **Start Lab** button. A countdown timer and a status circle icon will appear. Wait until the circle turns from amber to solid **Green** — this confirms all server clusters and security parameters are active.

3 Open the Console

Click the **AWS** text link directly to the left of the green status light. A clean, authenticated AWS Management Console session will open in a new browser tab.

Task 2 — Create a Standalone EBS Volume

Navigate to **EC2** → **Elastic Block Store** → **Volumes** using the top search bar. Click **Create volume** (upper-right) and configure the following settings:

Setting	Required Value	Why It Matters
Volume type	General Purpose SSD (gp3)	Default; within Academy sandbox cost limits
Size	10 GiB exactly	Larger sizes are blocked by SCPs
Availability Zone	Match your running EC2 instance (e.g., us-east-1a)	EBS volumes cannot attach across AZs



AZ Alignment is critical. An EBS volume functions like an internal hard disk over a low-latency local bus — it is structurally impossible to attach to an EC2 instance in a different Availability Zone. Always verify your AZ before clicking Create volume.

Task 3 — Attach the EBS Volume to Your EC2 Instance

→ Select the Volume

In the Volumes table, check the box next to your new 10 GiB volume. Confirm its lifecycle state reads **Available** before proceeding.

→ Confirm the Device Path

Leave the default device path (e.g., `/dev/sdf` or `/dev/xvdf`) unchanged. Click the orange **Attach volume** button.

→ Attach via Actions Menu

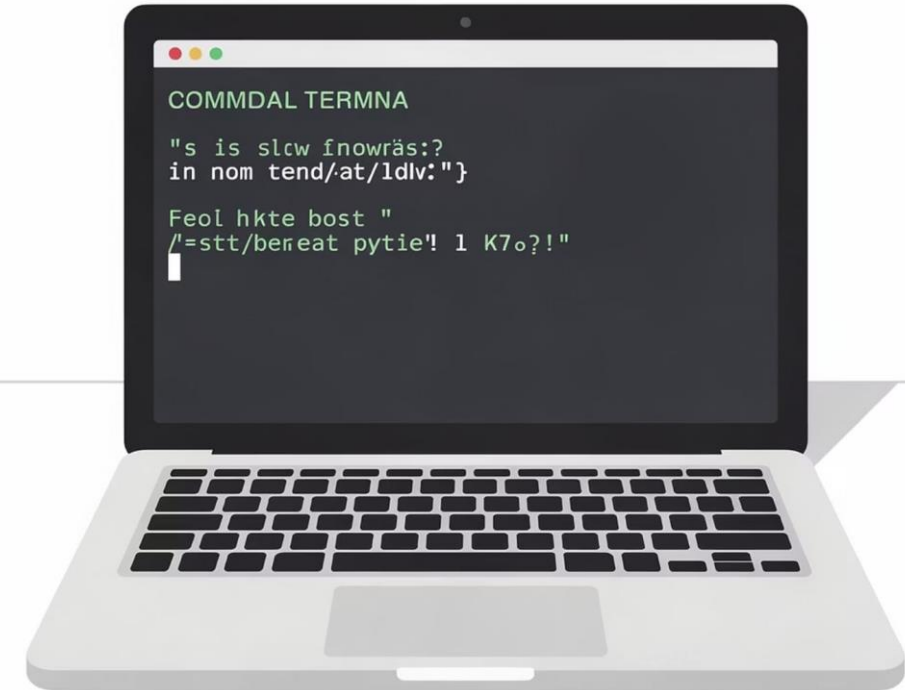
Click the **Actions** dropdown in the toolbar and choose **Attach volume**. In the Instance field, select the pre-provisioned virtual machine named **LabHost**.

→ Verify State Change

Watch the status column refresh. The volume state will change from *Available* to **In-Use**, confirming successful attachment.

Task 4 — Connect via CLI and Mount the Disk

Go to **EC2** → **Instances**, select **LabHost**, and click **Connect**. Choose the **EC2 Instance Connect** tab, confirm the username is `ec2-user`, and click **Connect**. A browser-based Linux terminal will open.



Task 4 — Mount Commands (Step-by-Step)

Run each command in sequence inside your EC2 Instance Connect terminal:

Step 1 — Verify Disk Attachment

```
lsblk
```

You should see a secondary device (`xvdf`, ~10G) listed alongside the root disk. This confirms the raw block device is recognized by the OS.

Step 2 — Format with XFS

```
sudo mkfs -t xfs /dev/xvdf
```

Applies an enterprise-standard XFS file system structure to the raw, empty disk sectors. **Do not skip this step** — mounting an unformatted volume will fail.

Steps 3-4 — Create Mount Point & Mount

```
sudo mkdir /mnt/data-store  
sudo mount /dev/xvdf /mnt/data-store
```

Creates the directory entry point and binds the formatted disk to it.

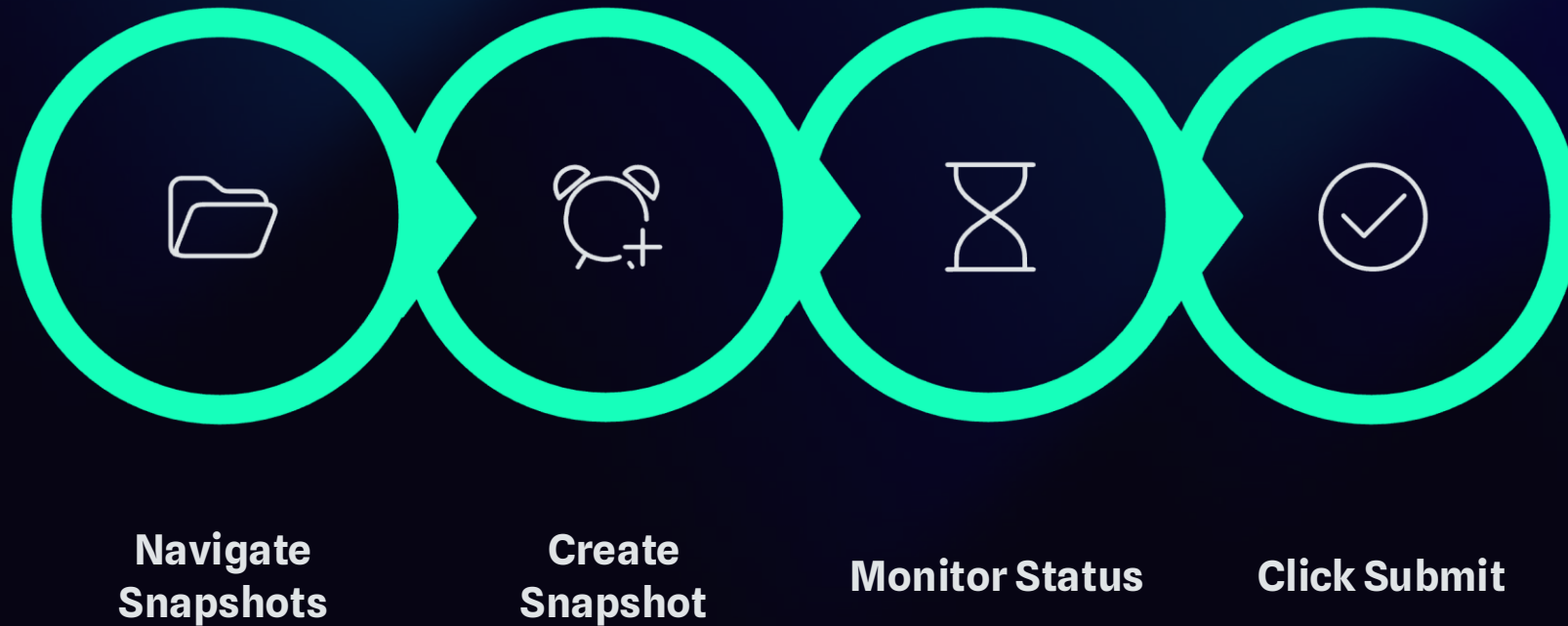
Step 5 — Verify & Write Test File

```
df -h  
echo "ITNW 1009 EBS Lab Verification Complete" | sudo  
tee /mnt/data-store/lab-check.txt
```

The `df -h` output must show `/dev/xvdf` mounted at `/mnt/data-store` with ~10G capacity.

Task 5 — Create an EBS Snapshot Backup

Snapshots are point-in-time backups stored in Amazon S3. They are essential for enterprise data durability. Follow these steps without closing your terminal tab:



In the Description field, enter exactly: **EBS Automated Class Backup Snapshot Target**. Once the status changes from *Pending* to **Completed**, return to the Academy dashboard and click **Submit / Check Progress** to lock in your grade points.

Part A — Troubleshooting Reference

Error Symptom	Root Cause	Remediation Steps
<i>"UnauthorizedOperation – You are not permitted to execute this action"</i> during volume creation	Selected a restricted volume type (e.g., io2) or a size greater than 10 GiB – both are blocked by Academy SCPs	Delete the draft. Restart Task 2 using gp3 type and exactly 10 GiB
Mount command prints: <i>"wrong fs type, bad option, bad superblock on /dev/xvdf"</i>	The formatting step (mkfs) was skipped – attempting to mount a raw, empty block device	Run sudo mkfs -t xfs /dev/xvdf first, then re-run the mount command

LAB PART B

Sandbox Scratch Build — S3, EC2 & Glacier Deep Archive

In Part B, you step into the role of a **Lead Cloud Systems Engineer**. You will build a complete cloud infrastructure environment from absolute scratch — no templates, no pre-made resources — inside the AWS Academy Learner Lab Sandbox.



Part B — Lab Objectives

Build a VPC

Construct an isolated virtual private network including custom subnets, an internet gateway, and routing tables.

Deploy EC2

Launch a cost-efficient t2.micro EC2 instance inside your custom VPC with a public IP address.

Security Group

Configure a custom security group to control inbound SSH access to your cloud resources.

S3 + Glacier

Create an S3 bucket and configure an automated lifecycle policy to archive objects to Glacier Deep Archive after 30 days.

Part B — Prerequisites & Budget Guardrails

Environment & Instance Rules

- **Environment:** AWS Academy Learner Lab Sandbox (unguided, no templates)
- **Instance Types:** Restrict all builds to **t2.micro** or **t3.micro** depending on your account's regional availability
- **AMI:** Use the default **Amazon Linux 2023 AMI** only

💰 \$100 Budget — Hard Limit

Your cumulative sandbox operations are strictly capped at **\$100 total**. Exceeding this limit will lock your account.

- Never leave unused Elastic IP allocations running
- Do not provision oversized EBS volumes
- **Always terminate resources** at the end of every session

Task 1 — Initialize the Sandbox Environment



Open the Learner Lab

Open your Canvas course portal and click the **AWS Academy Learner Lab Sandbox** module link.



Start the Lab

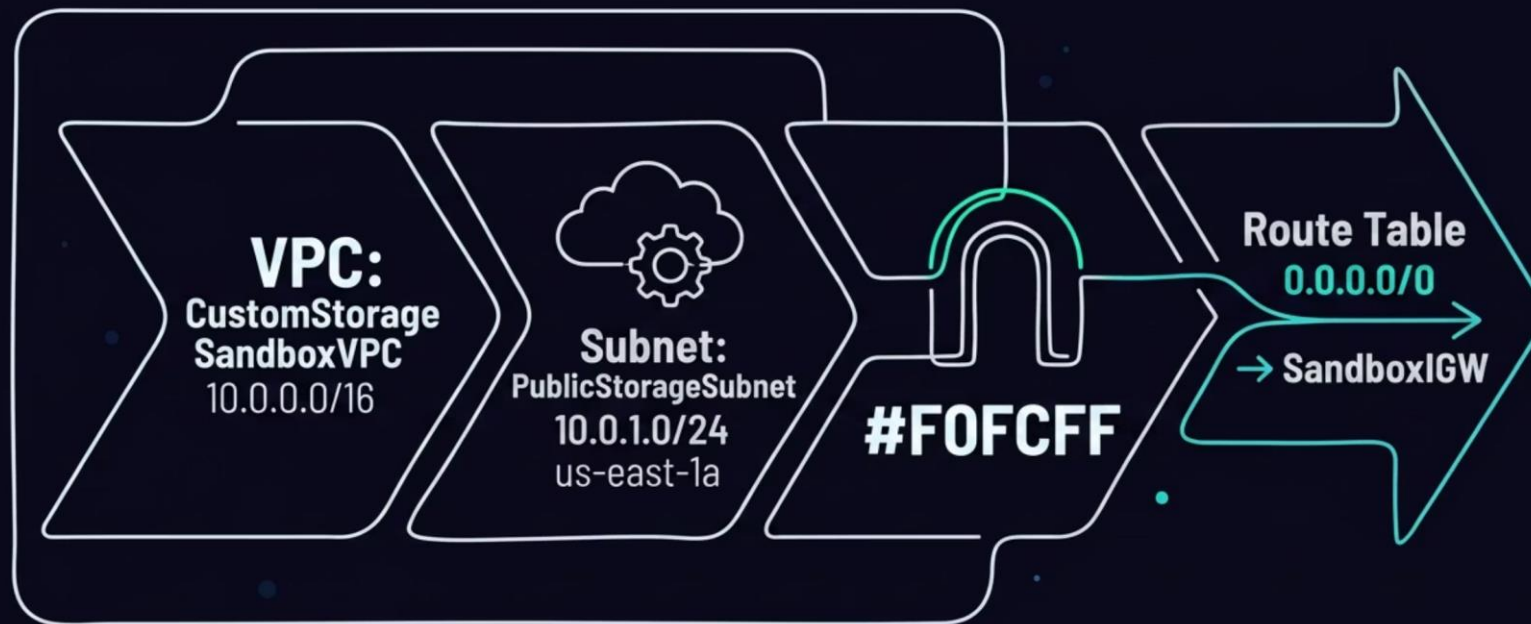
Click **Start Lab**. The engine will build a completely blank cloud sandbox account. Wait for the status circle to turn solid **Green**.



Verify Your Region

Click **AWS Console** to open a new tab. Confirm the top-right region dropdown reads **N. Virginia (us-east-1)** or your designated regional sector.

Task 2 — Build the Virtual Private Network (VPC)



Each component must be created in order — the subnet must reference the VPC, the IGW must be attached to the VPC, and the route table entry must point to the IGW before any EC2 instance can reach the internet.

Task 2 — VPC Configuration Values Reference

Resource	Name / Value	Notes
VPC	CustomStorageSandboxVPC	IPv4 CIDR: 10.0.0.0/16 · Select "VPC only"
Subnet	PublicStorageSubnet	AZ: us-east-1a · IPv4 CIDR: 10.0.1.0/24
Internet Gateway	SandboxIGW	After creation, use Actions → Attach to VPC
Route Table Rule	Destination: 0.0.0.0/0	Target: Internet Gateway → SandboxIGW

Task 3 — Configure the Security Group

Security groups act as virtual firewalls. Navigate to **VPC** → **Security** → **Security groups** and click **Create security group**.

Basic Identity

- **Name:** StorageServerSG
- **Description:** Authorize localized incoming SSH transport operations
- **VPC:** Select CustomStorageSandboxVPC (remove the default VPC)

Inbound Rule

- **Type:** SSH (auto-sets port to 22)
- **Source:** Anywhere-IPv4 (0.0.0.0/0)

This rule allows you to connect to your EC2 instance via the browser-based terminal during the lab session.

 In a real production environment, you would restrict SSH access to specific IP ranges rather than 0.0.0.0/0. For this lab sandbox, Anywhere-IPv4 is acceptable for simplicity.

Task 4 — Deploy a Cost-Efficient EC2 Instance

Navigate to **EC2** → **Instances** → **Launch instances**. Configure the instance as follows:

Instance Settings

- **Name:** SandboxStorageHost
- **AMI:** Amazon Linux 2023 AMI (default)
- **Instance type:** t2.micro or t3.micro
- **Key pair:** Select vockey (pre-built sandbox key)

Network Settings (click Edit)

- **VPC:** CustomStorageSandboxVPC
- **Subnet:** PublicStorageSubnet
- **Auto-assign public IP:** Enable
- **Security group:** Select existing → StorageServerSG

Click the orange **Launch instance** button. Your EC2 instance will begin initializing within your custom VPC.

LAB PART B – TASK 5

Provision Your S3 Bucket

Amazon S3 stores objects (files) in globally unique named containers called buckets. In the console search bar, type **S3** and click to open. Then click **Create bucket**.



Task 5 — S3 Bucket Configuration

1

Name Your Bucket (Critical)

S3 bucket names are **globally unique** across all AWS accounts worldwide. Use this exact format:

`itnw1009-sandboxoutage-yourname-2026`

All lowercase, no spaces or special symbols.

2

Object Ownership

Leave set to **ACLs disabled (recommended)**. This is the most secure default configuration for access control.

3

Block Public Access

Keep **all checkboxes checked**. This prevents any accidental public internet exposure of your stored files.

4

Create the Bucket

Scroll to the bottom and click **Create bucket**. Click your bucket name in the table to enter its dashboard panel.

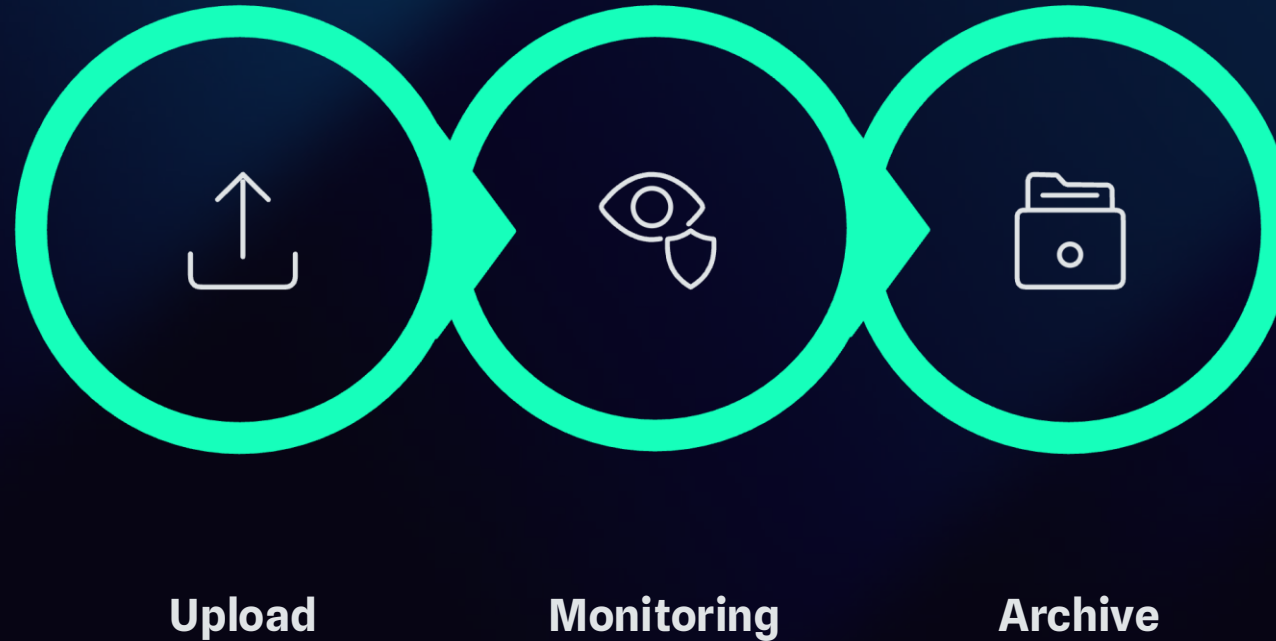
Task 5 — Configure the Glacier Deep Archive Lifecycle Policy

Inside your bucket, click the **Management** tab → **Create lifecycle rule**. Fill in the configuration fields:

Field	Required Value
Lifecycle rule name	AutomatedGlacierDeepArchiveTransitionPolicy
Rule scope	Apply to all objects in the bucket (check the confirmation warning)
Lifecycle rule action	Move current versions of objects between storage classes
Storage class transition	Glacier Deep Archive
Days after object creation	30

Click **Create rule**. Your automated archival pipeline is now active.

How the Glacier Lifecycle Policy Works



This programmatic policy continuously queries S3 object metadata. Exactly 30 calendar days after any object's initial upload timestamp, the automated engine strips it from high-cost active S3 storage and archives it in the rock-bottom-cost Glacier Deep Archive compliance vault — with zero manual intervention required.

S3 Storage Classes — Cost vs. Access Speed

S3 Standard

Highest Cost
Active Data

Millisecond access.

S3 Standard-IA

Lower Cost

Infrequent access, hours retrieval.

S3 Glacier Instant Retrieval

Millisecond Retrieval
Archival Cost

S3 Glacier Deep Archive

Lowest Cost

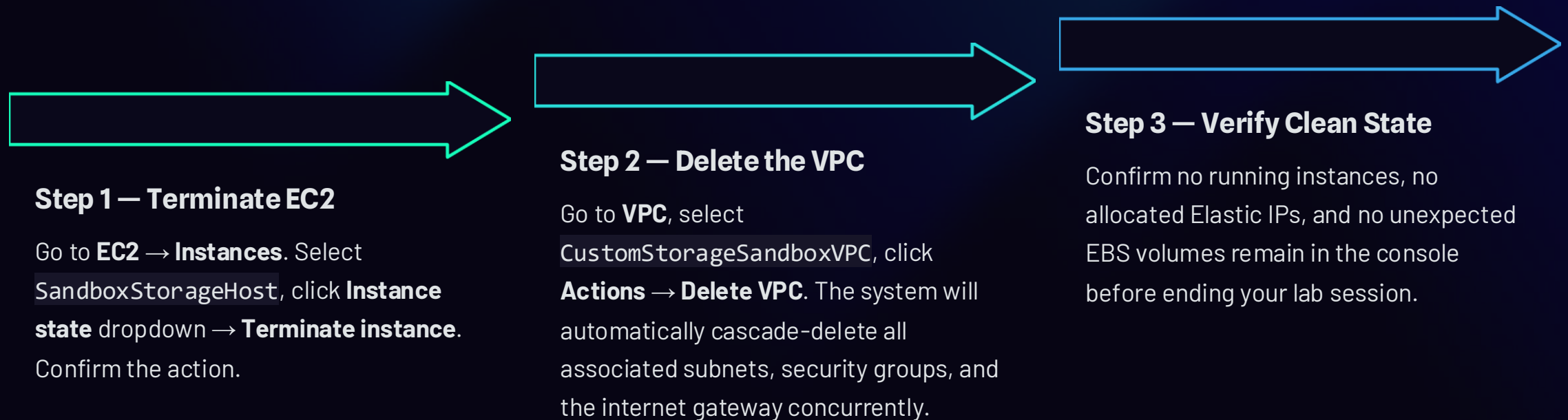
12-hour retrieval, long-term compliance.

Part B — Troubleshooting Reference

Error Symptom	Root Cause	Remediation Steps
S3 creation fails: <i>"Bucket name already exists"</i>	Your chosen name is already in use by another AWS account globally — bucket names must be universally unique	Return to Task 5, Step 3. Append random digits or your student ID to the bucket name string and resubmit
EC2 instance boots but terminal returns a <i>connection timeout</i> error	The VPC route table is missing the <code>0.0.0.0/0 → SandboxIGW</code> entry, or the subnet is not associated with the route table	Revisit Task 2, Steps 15–18. Verify the destination <code>0.0.0.0/0</code> targets <code>SandboxIGW</code> and confirm subnet association under the Subnet Associations tab

Final Environment Teardown — Protect Your \$100 Budget

⚠ You MUST complete this teardown at the end of every lab session. Leaving EC2 instances, Elastic IPs, or large EBS volumes running will consume your \$100 sandbox budget and may trigger unexpected infrastructure charges during grading breaks.



Key Concepts Review — Storage Services Compared



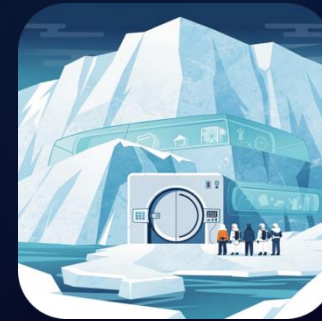
Amazon EBS

Persistent **block storage** attached to a single EC2 instance. Behaves like a physical internal hard disk. Must be in the same AZ as its instance. Ideal for OS volumes, databases, and low-latency I/O workloads.



Amazon S3

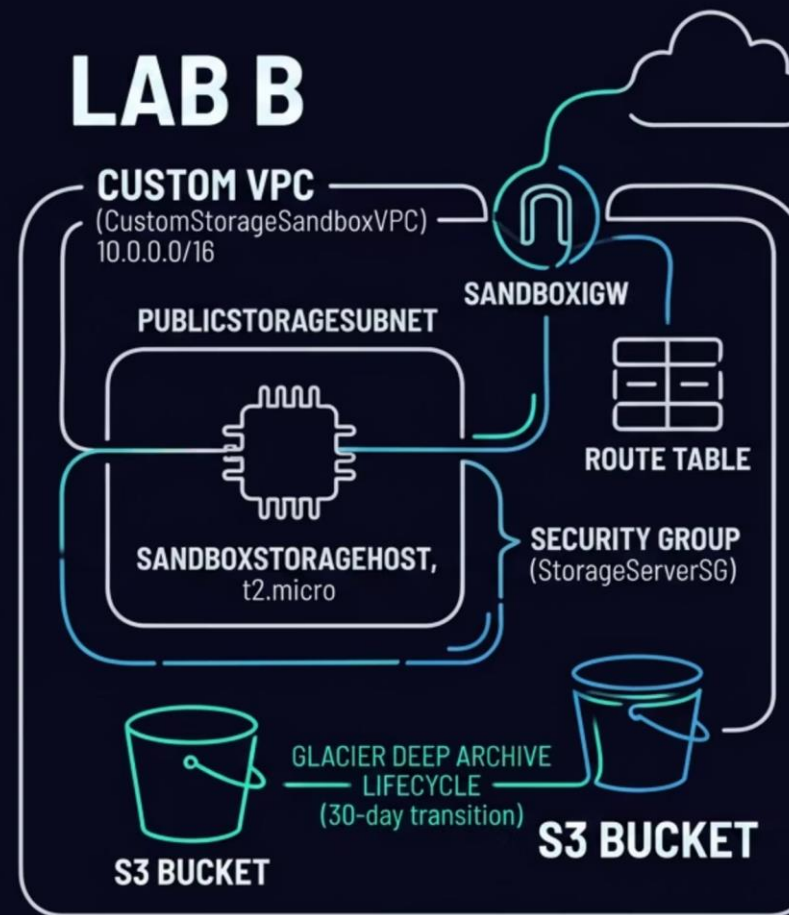
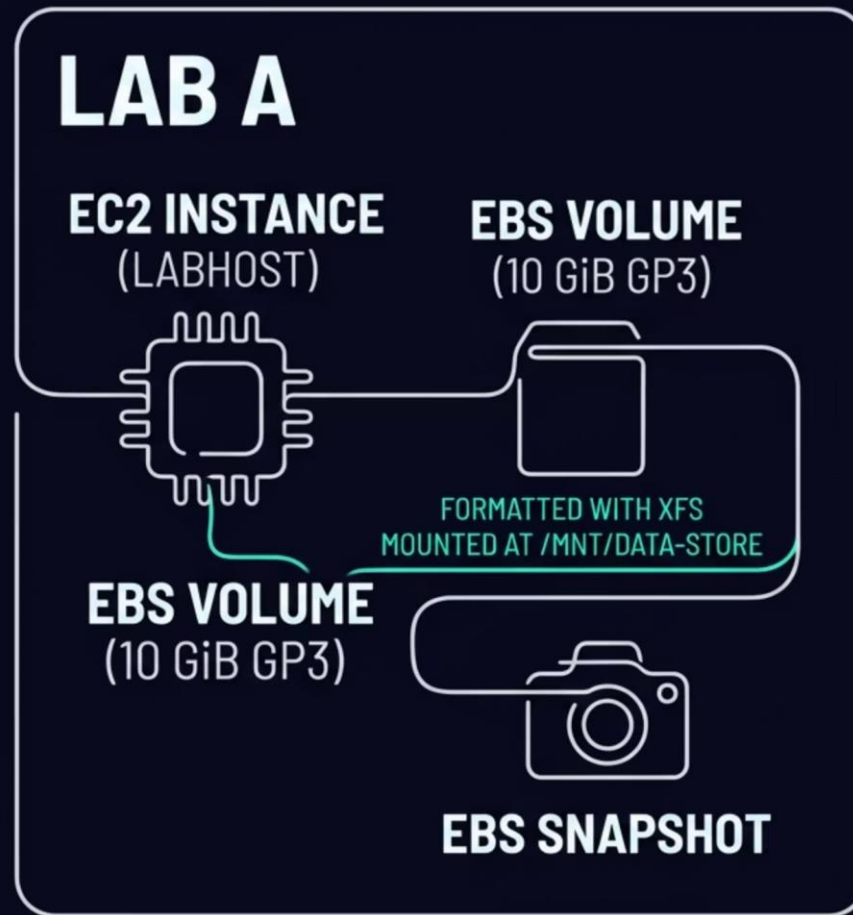
Scalable **object storage** accessible from anywhere. Stores files (objects) inside globally-named buckets. Ideal for backups, static websites, data lakes, and media delivery.



S3 Glacier Deep Archive

The lowest-cost AWS storage class for **long-term compliance archiving**. Retrieval takes up to 12 hours. Best for regulatory records, audit logs, and data retained for 7–10+ years.

What You Built Today — Architecture Summary



Day 7 Complete — Well Done!

You have successfully completed both AWS Academy lab segments for Day 7. You now have hands-on experience with the core building blocks of cloud storage architecture used by enterprises worldwide.

Part A Achieved

Created, attached, formatted, mounted, and snapshotted an EBS volume on a live EC2 instance using the Linux CLI.

Part B Achieved

Built a complete VPC, deployed an EC2 instance, created an S3 bucket, and configured an automated Glacier Deep Archive lifecycle policy.

Next Steps

Review the AWS Documentation for EBS Snapshots and S3 Lifecycle Policies. Complete any remaining Canvas quiz items before your next session.

